INCH-POUND
MIL-DTL-24643/50F
27 November 2023
SUPERSEDING
MIL-DTL-24643/50E
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 1,000 VOLTS, TYPE LSFNW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation (see table I for thickness). Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, the insulation color shall be black.
Fourth	-	Four conductors cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm, well-rounded assembly. Cabling sequence shall be consecutive.
Fifth	-	An optional binder.
Sixth	-	Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/50	Type and size	Conductor size (AWG)	Insulation thickness min. avg. (inch)	Cable jacket thickness min. avg. (inch)	Overall diameter (inches)	
					min.	max.
-01UN	LSFNW-3	16 (class B)	0.030	0.040	0.413	0.447
-02UN	LSFNW-4	14 (class B)	0.030	0.040	0.475	0.513
-03UN	LSFNW-9	10 (class B)	0.030	0.040	0.583	0.630
-04UN	LSFNW-23	7 (class B)	0.045	0.055	0.768	0.830
-05UN	LSFNW-50	3 (class C)	0.045	0.060	0.993	1.074
-06UN	LSFNW-75	1 (class C)	0.055	0.070	1.232	1.332
-07UN	LSFNW-100	0 (class D)	0.055	0.075	1.337	1.446
-08UN	LSFNW-150	000 (class D)	0.055	0.075	1.565	1.692
-09UN	LSFNW-200	0000 (class D)	0.055	0.075	1.698	1.836

TABLE II. Electrical details.

Type and size	Conductor resistance (ohms)	Insulation resistance (megohms)	Voltage withstand min. (volts)	Accelerated service loading current (amperes)
LSFNW-3	4.3	6,000	4,000	N/A
LSFNW-4	2.68	5,000	4,000	N/A
LSFNW-9	1.08	3,800	4,000	N/A
LSFNW-23	0.543	3,700	5,500	115
LSFNW-50	0.210	2,600	6,000	185
LSFNW-75	0.134	2,600	6,000	250
LSFNW-100	0.106	2,400	6,000	295
LSFNW-150	0.0674	1,800	6,000	400
LSFNW-200	0.0530	1,500	6,500	465

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.) — See table II

Voltage withstand (volts, root mean square, min.)

Conductor to conductor — See table II

Insulation resistance (megohms/1,000 feet, min.)

Conductor to conductor — See table II

Conductor continuity — No failure

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Group A:	
Visual and dimensional	No failure
Group B:	
Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation), method 1 only (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Accelerated service – sizes 23 and larger (see table II for load current)	No failure

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Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSFNW	3	2,500
LSFNW	4 and 9	2,000
LSFNW	23 and 50	1,500
LSFNW	75 through 150	1,000
LSFNW	200	500

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-066)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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